

Complete Experiment Log Bundle Report

Included files:

- reports/complete_experiment_log_bundle.md
- reports/metric_trajectory_plot.svg
- reports/keep_discard_crash_summary.md
- reports/best_result_vs_baseline.md
- reports/what_actually_worked_memo.md

Complete Experiment Log Bundle

Complete Experiment Log Bundle

Bundle Contents

- Source log: `results/historical\_experiment\_log.csv`
- Latest matrix: `reports/experiment\_result\_matrix.csv`
- Result matrix: `reports/experiment\_result\_matrix.md`
- Metric plots: `reports/metric\_trajectory\_plot.svg`, `reports/metric\_over\_time\_plot.svg`
- Keep/discard/crash summary: `reports/keep\_discard\_crash\_summary.md`
- Best result comparison: `reports/best\_result\_vs\_baseline.md`

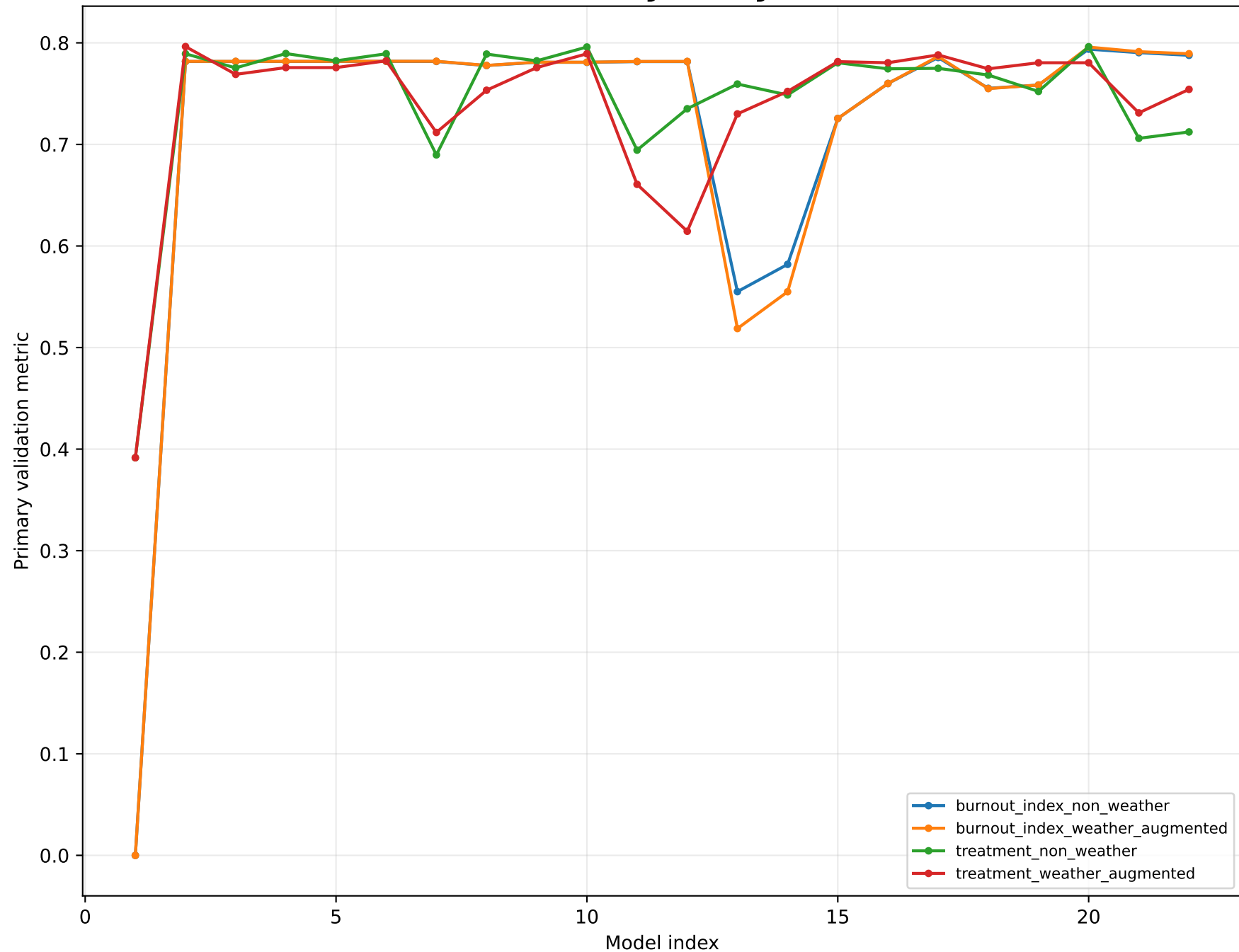
Latest Best Models

run_id	target	task_type	model	primary_metric	accuracy	f1_weighted	rmse	mae	r2
---	---	---	---	---	---	---	---	---	---
burnout_index_non_weather	burnout_index	regression	HistGradientBoostingReg	0.7938	n/a	n/a	0.8831	0.7084	0.7938
burnout_index_weather_augmented	burnout_index	regression	HistGradientBoostingReg	0.7958	n/a	n/a	0.8788	0.7054	0.7958
treatment_non_weather	sought_treatment	classification	HistGradientBoosting	0.7962	0.7959	0.7962	n/a	n/a	n/a
treatment_weather_augmented	sought_treatment	classification	Logistic_C0.1	0.7963	0.7959	0.7963	n/a	n/a	n/a

Reproducibility Rules

- Fixed random seed: `42`.
- Latest runs use validation metrics only.
- Historical runs are appended to `results/historical\_experiment\_log.csv`.

Metric Trajectory Plot



Classification uses weighted F1. Burnout-index regression uses R-squared.

Keep / Discard / Crash Summary

Keep / Discard / Crash Summary

run_id	model	primary_metric	decision
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treatment_non_weather	DummyMostFrequent	0.3915	Discard
treatment_non_weather	Logistic_C0.1	0.7893	Discard
treatment_non_weather	Logistic_C1	0.7753	Discard
treatment_non_weather	Logistic_C10	0.7895	Discard
treatment_non_weather	Ridge_alpha1	0.7823	Discard
treatment_non_weather	Ridge_alpha10	0.7893	Discard
treatment_non_weather	SGDLogistic_alpha0001	0.6897	Discard
treatment_non_weather	SGDLogistic_alpha001	0.7890	Discard
treatment_non_weather	LinearSVC_C0.5	0.7823	Discard
treatment_non_weather	LinearSVC_C1	0.7959	Discard
treatment_non_weather	KNN_k5	0.6944	Discard
treatment_non_weather	KNN_k15	0.7351	Discard
treatment_non_weather	DecisionTree_depth4	0.7595	Discard
treatment_non_weather	DecisionTree_depth8	0.7487	Discard
treatment_non_weather	RandomForest_depth8	0.7804	Discard
treatment_non_weather	RandomForest_full	0.7745	Discard
treatment_non_weather	ExtraTrees_depth8	0.7749	Discard
treatment_non_weather	ExtraTrees_full	0.7683	Discard
treatment_non_weather	GradientBoosting_lr005	0.7522	Discard
treatment_non_weather	HistGradientBoosting	0.7962	Keep
treatment_non_weather	NeuralNet_MLP_32	0.7060	Discard
treatment_non_weather	NeuralNet_MLP_64_32	0.7123	Discard
treatment_weather_augmented	DummyMostFrequent	0.3915	Discard
treatment_weather_augmented	Logistic_C0.1	0.7963	Keep
treatment_weather_augmented	Logistic_C1	0.7690	Discard
treatment_weather_augmented	Logistic_C10	0.7757	Discard
treatment_weather_augmented	Ridge_alpha1	0.7757	Discard
treatment_weather_augmented	Ridge_alpha10	0.7823	Discard
treatment_weather_augmented	SGDLogistic_alpha0001	0.7118	Discard
treatment_weather_augmented	SGDLogistic_alpha001	0.7534	Discard
treatment_weather_augmented	LinearSVC_C0.5	0.7757	Discard
treatment_weather_augmented	LinearSVC_C1	0.7893	Discard
treatment_weather_augmented	KNN_k5	0.6607	Discard
treatment_weather_augmented	KNN_k15	0.6145	Discard

Keep / Discard / Crash Summary continued

treatment_weather_augmented	DecisionTree_depth4	0.7301	Discard	
treatment_weather_augmented	DecisionTree_depth8	0.7522	Discard	
treatment_weather_augmented	RandomForest_depth8	0.7815	Discard	
treatment_weather_augmented	RandomForest_full	0.7804	Discard	
treatment_weather_augmented	ExtraTrees_depth8	0.7881	Discard	
treatment_weather_augmented	ExtraTrees_full	0.7745	Discard	
treatment_weather_augmented	GradientBoosting_lr005	0.7804	Discard	
treatment_weather_augmented	HistGradientBoosting	0.7804	Discard	
treatment_weather_augmented	NeuralNet_MLP_32	0.7311	Discard	
treatment_weather_augmented	NeuralNet_MLP_64_32	0.7542	Discard	
burnout_index_non_weather	DummyMean	-0.0001	Discard	
burnout_index_non_weather	LinearRegression	0.7816	Discard	
burnout_index_non_weather	Ridge_alpha1	0.7816	Discard	
burnout_index_non_weather	Ridge_alpha10	0.7816	Discard	
burnout_index_non_weather	Lasso_alpha0001	0.7817	Discard	
burnout_index_non_weather	Lasso_alpha001	0.7818	Discard	
burnout_index_non_weather	ElasticNet_alpha001	0.7817	Discard	
burnout_index_non_weather	ElasticNet_alpha01	0.7777	Discard	
burnout_index_non_weather	SGDReg_alpha0001	0.7809	Discard	
burnout_index_non_weather	SGDReg_alpha001	0.7808	Discard	
burnout_index_non_weather	LinearSVR_C0.5	0.7816	Discard	
burnout_index_non_weather	LinearSVR_C1	0.7816	Discard	
burnout_index_non_weather	KNNReg_k5	0.5551	Discard	
burnout_index_non_weather	KNNReg_k15	0.5819	Discard	
burnout_index_non_weather	DecisionTreeReg_depth6	0.7255	Discard	
burnout_index_non_weather	RandomForestReg_depth8	0.7600	Discard	
burnout_index_non_weather	RandomForestReg_full	0.7856	Discard	
burnout_index_non_weather	ExtraTreesReg_depth8	0.7552	Discard	
burnout_index_non_weather	GradientBoostingReg	0.7585	Discard	
burnout_index_non_weather	HistGradientBoostingReg	0.7938	Keep	
burnout_index_non_weather	NeuralNetReg_MLP_32	0.7904	Discard	
burnout_index_non_weather	NeuralNetReg_MLP_64_32	0.7878	Discard	
burnout_index_weather_augmented	DummyMean	-0.0001	Discard	
burnout_index_weather_augmented	LinearRegression	0.7818	Discard	
burnout_index_weather_augmented	Ridge_alpha1	0.7818	Discard	
burnout_index_weather_augmented	Ridge_alpha10	0.7818	Discard	
burnout_index_weather_augmented	Lasso_alpha0001	0.7818	Discard	
burnout_index_weather_augmented	Lasso_alpha001	0.7819	Discard	

Keep / Discard / Crash Summary continued

burnout_index_weather_augmented	ElasticNet_alpha001	0.7819	Discard	
burnout_index_weather_augmented	ElasticNet_alpha01	0.7778	Discard	
burnout_index_weather_augmented	SGDReg_alpha0001	0.7810	Discard	
burnout_index_weather_augmented	SGDReg_alpha001	0.7810	Discard	
burnout_index_weather_augmented	LinearSVR_C0.5	0.7817	Discard	
burnout_index_weather_augmented	LinearSVR_C1	0.7817	Discard	
burnout_index_weather_augmented	KNNReg_k5	0.5187	Discard	
burnout_index_weather_augmented	KNNReg_k15	0.5549	Discard	
burnout_index_weather_augmented	DecisionTreeReg_depth6	0.7256	Discard	
burnout_index_weather_augmented	RandomForestReg_depth8	0.7600	Discard	
burnout_index_weather_augmented	RandomForestReg_full	0.7866	Discard	
burnout_index_weather_augmented	ExtraTreesReg_depth8	0.7550	Discard	
burnout_index_weather_augmented	GradientBoostingReg	0.7585	Discard	
burnout_index_weather_augmented	HistGradientBoostingReg	0.7958	Keep	
burnout_index_weather_augmented	NeuralNetReg_MLP_32	0.7913	Discard	
burnout_index_weather_augmented	NeuralNetReg_MLP_64_32	0.7893	Discard	

Best Result vs. Baseline

Best Result vs. Baseline

run_id	baseline_model	best_model	baseline_primary	best_primary	delta
---	---	---	---	---	---
burnout_index_non_weather	DummyMean	HistGradientBoostingReg	-0.0001	0.7938	0.7939
burnout_index_weather_augmented	DummyMean	HistGradientBoostingReg	-0.0001	0.7958	0.7959
treatment_non_weather	DummyMostFrequent	HistGradientBoosting	0.3915	0.7962	0.4047
treatment_weather_augmented	DummyMostFrequent	Logistic_C0.1	0.3915	0.7963	0.4048

Decision Rule

Keep the best model only when it improves the run's primary validation metric over the dummy baseline.

What Actually Worked Memo

What Actually Worked Memo

- `burnout\_index\_non\_weather`: `HistGradientBoostingReg` was best with primary metric 0.7938.
- `burnout\_index\_weather\_augmented`: `HistGradientBoostingReg` was best with primary metric 0.7958.
- `treatment\_non\_weather`: `HistGradientBoosting` was best with primary metric 0.7962.
- `treatment\_weather\_augmented`: `Logistic\_C0.1` was best with primary metric 0.7963.

Weather improves a run only if the best weather-augmented primary metric exceeds the best non-weather primary metric for the same target.